DHAROHAR: Safeguarding India's Wisdom with Digital Sovereignty

Master Project Documentation (PRD + SRS + Architecture)

1. Product Requirements Document (PRD)

1.1 Executive Summary

India possesses massive "intangible capital"—tribal medicinal knowledge, intricate handloom techniques, and oral folklore. However, this capital is currently illiquid and vulnerable. It is either stolen (bio-piracy), counterfeited (powerloom fakes), or lost to time. **Dharohar** is a digital infrastructure that transforms this unstructured heritage into verifiable, defensible, and monetizable economic assets. We do not just archive culture; we tokenize it.

1.2 Problem Statement

- **The Documentation Gap:** 80% of indigenous knowledge is oral. Under patent law, oral knowledge is rarely accepted as "Prior Art," making it easy for corporations to patent traditional remedies (e.g., the Turmeric patent case).
- **The Authenticity Crisis:** Artisans cannot mathematically prove the difference between a 30-day handwoven saree and a 30-minute machine-made copy, leading to price collapse.
- **Economic Exclusion:** Communities are "Asset Rich but Cash Poor" because they lack the legal and digital tools to license their knowledge.

1.3 Proposed Solution

A "Digital Sovereignty" platform featuring three core engines:

1. **Dharohar-Bio:** Converts oral ethnobotany into legal "Prior Art Dossiers" and licensing contracts.
2. **Dharohar-Craft:** Uses Computer Vision to create "Digital Twins" of physical handicrafts, verifying authenticity via weave topology.
3. **Dharohar-Sonic:** Fingerprints folk music to manage copyrights and royalties.

1.4 User Personas

- **The Custodian (Creator):** Tribal elders, master weavers, or Gram Sabhas. They have low digital literacy and require voice-first interfaces.
- **The Certifier (Verifier):** Biodiversity Management Committees (BMCs), Textile Ministry officers, or Ethnobotanists who validate claims.
- **The Licensee (Buyer):** Pharma R&D units, Ethical Fashion Houses, and Academic Researchers.

1.5 Success Metrics (KPIs)

- **Defensibility:** Number of "Prior Art" claims successfully registered with the National Biodiversity Authority.
- **Economic Velocity:** Total royalty volume flowed back to community wallets.
- **Accuracy:** False Positive Rate < 2% for AI counterfeit detection.

2. Software Requirements Specification (SRS)

2.1 Functional Requirements

Module A: Ingestion & Digitization

- **Multimodal Capture:** System must accept Audio (Voice notes in dialects), Video (Macro-lens footage of crafts), and Image data.
- **Geospatial Stamping:** App must capture immutable GPS coordinates at the moment of recording to establish "Geographical Indication" (GI).
- **Offline Mode:** Data must buffer locally on the device and sync via MQTT when connectivity is restored.

Module B: AI Processing & Validation

- **Dialect-to-Data (Bio):** System must transcribe specific tribal dialects (e.g., Gondi, Bhili) and perform Named Entity Recognition (NER) to map local plant names to scientific taxonomy.
- **Pattern Forensics (Craft):** Computer Vision models must analyze video frames to detect "human error signatures" in thread tension, distinguishing handloom from powerloom.
- **Prior Art Generation:** The system must auto-generate a PDF compliant with "Form I" of the Biological Diversity Act, 2002.

Module C: Asset Minting & Governance

- **Verifiable Credentials:** Upon expert approval, the system must mint a cryptographic token (Soulbound Token or VC) representing the asset.
- **Role-Based Access (RBA):**
  - *Public View:* Metadata only (e.g., "Herb treats fever").
  - *Private View:* Proprietary knowledge (e.g., "Boil root at 80°C for 20 mins"). Accessible only after Smart Contract execution.

2.2 Non-Functional Requirements

- **Data Sovereignty:** All sensitive data must be encrypted at rest using keys managed ideally by the community trust (or a proxy).
- **Latency:** Inference for counterfeit detection must occur under 3 seconds on standard 4G networks.

- **Auditability:** Every view, download, and license purchase must be recorded on an immutable ledger.

3. Technical Architecture & AWS Stack

This architecture prioritizes "Serverless" for scale and "Managed Services" for AI depth.

3.1 The "Intellectual" Stack

Layer	Component	AWS Service / Technology	Justification
Edge / Client	Front-End	**React Native** (Mobile) + **Amplify**	Cross-platform, offline-first capabilities.
Ingestion	API Gateway	**Amazon API Gateway**	Handles burst traffic from village camps.
Compute	Logic	**AWS Lambda** (Python/Node.js)	Event-driven architecture; zero idle cost.
Storage	Media	**Amazon S3** (Intelligent-Tiering)	Cost-effective storage for large video archives.
Storage	Metadata	**Amazon DynamoDB**	Millisecond latency for querying asset attributes.
AI (Vision)	Craft Forensics	**Amazon Rekognition (Custom Labels)**	Trained on "Handloom vs. Powerloom" datasets.
AI (Language)	Bio-Extraction	**Amazon Bedrock (Claude 3.5 Sonnet)**	Superior reasoning for translating dialect to legal text.
AI (Audio)	Transcription	**Amazon Transcribe** + **Custom Vocab**	Fine-tuned for Indian rural accents/dialects.
Blockchain	Ledger	**Amazon Managed Blockchain (Access)**	Connects to Polygon/Ethereum for public asset minting.
Security	Identity	**Amazon Cognito**	Granular RBA (Creators vs. Verifiers).
Compliance	Audit Logs	**Amazon QLDB**	Cryptographically verifiable log of all data access.

3.2 Data Flow Logic

1. **Ingestion:** User records video -> App uploads to S3 -> Triggers Lambda.
2. **Processing:**
    - *If Video:* Lambda triggers **Rekognition**. Rekognition returns "Handloom Confidence Score: 98%".

- *If Audio:* Lambda triggers **Transcribe** -> Output text sent to **Bedrock**. Bedrock extracts entities (Plant: *Terminalia arjuna*, Use: *Cardiac Health*).

3. **Verification:** Metadata + AI Score stored in **DynamoDB**. "Pending" flag set.
4. **Minting:** Verifier approves on Dashboard -> Lambda calls **Smart Contract** via **Managed Blockchain** -> Asset Token minted.
5. **Licensing:** Buyer pays via Gateway -> Smart Contract splits funds -> **QLDB** records the transfer of rights.

4. Implementation Roadmap

Phase 1: The "Bio-Shield" MVP (Weeks 1-4)

- **Goal:** Protect Tribal Remedies.
- **Build:**
  - Mobile App with Voice Recorder and GPS.
  - AWS Bedrock pipeline to convert Voice-to-Legal-PDF.
  - Simple S3 storage structure.
- **Demo:** Speak a remedy in broken Hindi; system generates a downloadable "Prior Art" PDF.

Phase 2: The "Visual Truth" Update (Weeks 5-8)

- **Goal:** Verify Handicrafts.
- **Build:**
  - Data collection: 50 videos of Weavers vs. Machines.
  - Train **Rekognition Custom Labels** model.
  - Integrate Verification Dashboard for experts.
- **Demo:** Point camera at a fabric; App flags "Authentic" or "Counterfeit."

Phase 3: The "Sovereign Market" (Weeks 9-12)

- **Goal:** Monetization.
- **Build:**
  - Deploy Smart Contracts (Polygon/Ethereum) via Amazon Managed Blockchain.
  - Marketplace Web Portal for buyers.
  - Wallet integration (UPI to Crypto bridge).
- **Demo:** Purchase a license for a remedy; show the transaction split on the blockchain explorer.

4.1 Future Scope

- **IoT Integration:** Smart Sensors on looms (using **AWS IoT Core**) to track weave time actively.

- **Genetic Mapping:** Storing DNA sequences of plants alongside the traditional knowledge.